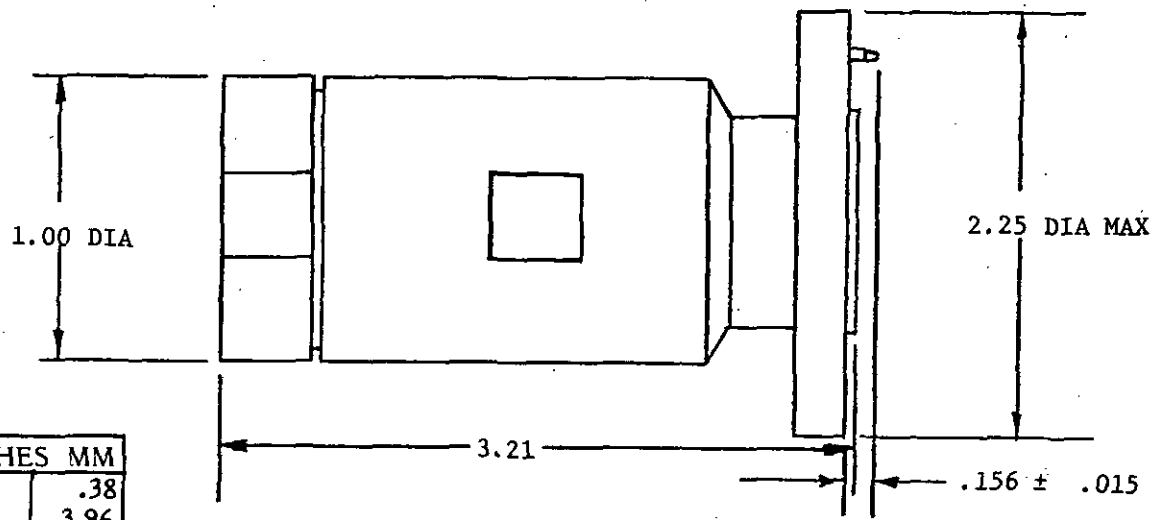
MILITARY SPECIFICATION SHEET

CONNECTORS, COAXIAL, RADIO FREQUENCY SERIES EIA FLANGE (SEMIRIGID CABLE, CLASS 2) (FOR USE WITH MIL-C-28830/3)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-39012.



INCHES	MM
.015	.38
.156	3.96
1.00	25.40
2.25	57.15
3.21	81.53

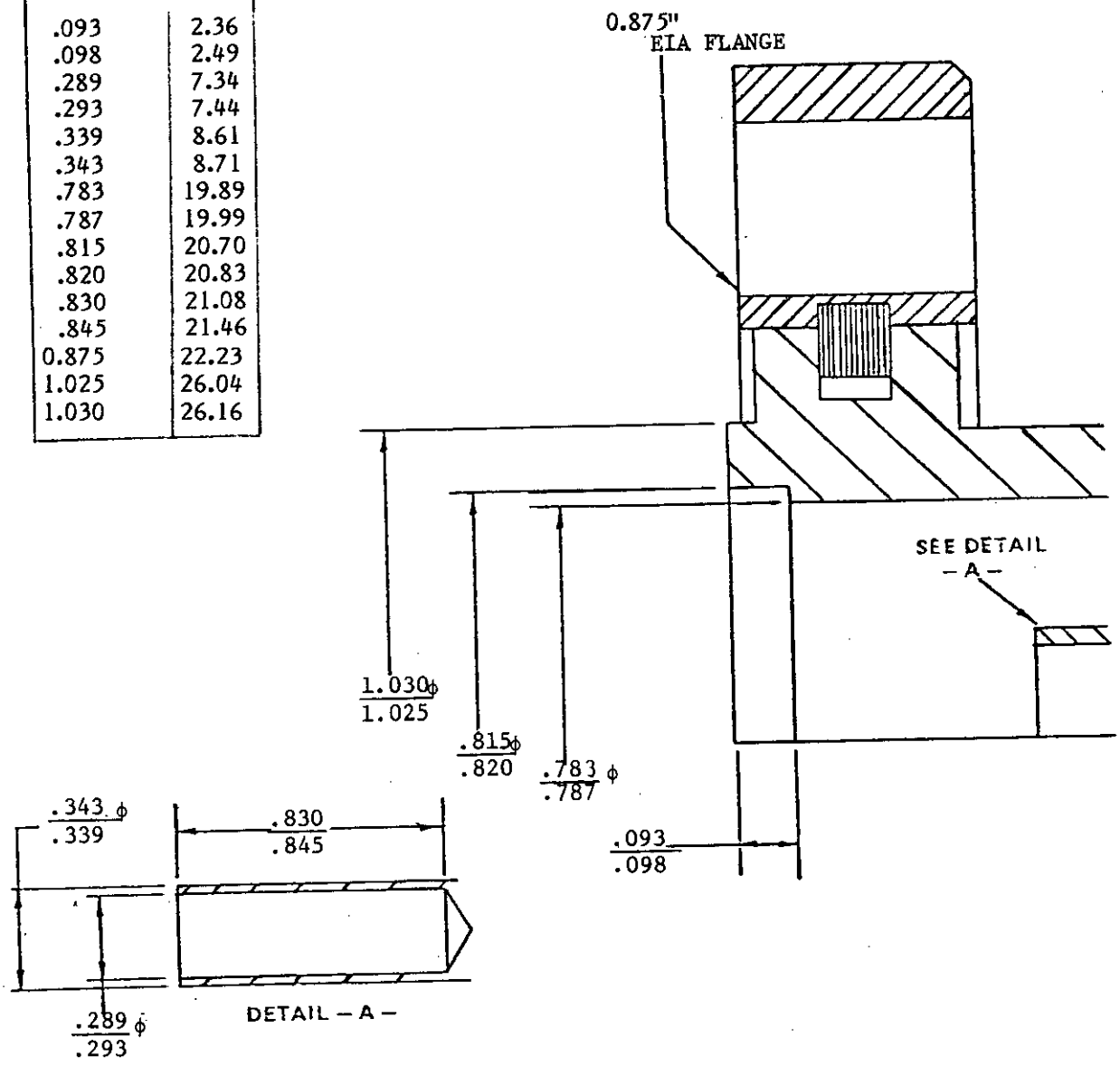
NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
3. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. General configuration.

FSC 5935

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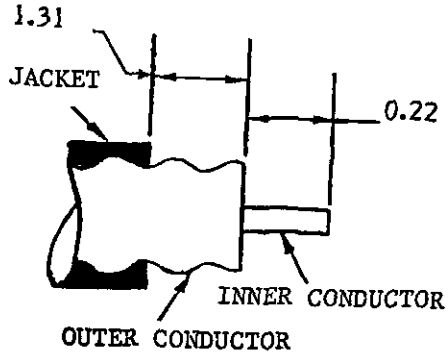
INCHES	MM
.093	2.36
.098	2.49
.289	7.34
.293	7.44
.339	8.61
.343	8.71
.783	19.89
.787	19.99
.815	20.70
.820	20.83
.830	21.08
.845	21.46
0.875	22.23
1.025	26.04
1.030	26.16



NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Mating dimensions for pin terminations.

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INCHES	MM
0.38	10
1.31	33

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Cable outer conductor to be cut at crest only.

FIGURE 3. Cable stripping dimensions for connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms
Frequency range: 0 to 8.1 GHz
Voltage rating: 1500 volts

Temperature range: $-40^{\circ}C$ to $+100^{\circ}C$
Peak RF power: 19 kW

REQUIREMENTS:

Dimensions and configuration: See figure 1.
Force to engage and disengage:
Longitudinal force - Not applicable
Torque - Not applicable

Coupling proof torque: Not applicable
Inspection conditions: Not applicable
Mating characteristics: See figure 2 for dimensions.
Hermetic seal: Not applicable.
Leakage (pressurized connectors): Not applicable.
Insulation resistance: Method 302, test condition B, MIL-STD-202.
5,000 megohms minimum.
Center contact retention: 6 pounds minimum.
Radial torque: Not applicable.
Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.
Voltage standing wave ratio (VSWR) per connector: 0.5 to 8.0 GHz - 1.2:1

Swept frequency VSWR test setup:
Item 6 - VSWR shall be less than 1.008 + .002 f (f in GHz).
Item 16 - VSWR shall be less than 1.008 + .002 f (f in GHz).
Second step of VSWR checkout procedure - VSWR shall be less than 1.012 + .005 f (f in GHz)
Group B inspection - 1.024 + .007
Qualification and group C inspection shall not exceed 1.08

Connector durability: 50 cycles minimum at 2 cycles per minute maximum

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Contact resistance: In milliohms maximum.

	Initial	After environment
Inner contact:	1.0	1.0
Outer contact:	0.2	0.2

Dielectric withstanding voltage at sea level: 4000 volts
Method 301 of MIL-STD-202

Vibration, high frequency: Method 204, test condition D, MIL-STD-202. Cable clamps shall be mounted on the vibration table 6 inches minimum from the mating face of the connector pair.

Shock: Method 213, test condition 1, MIL-STD-202.

Temperature cycling: Method 102, test condition C, MIL-STD-202, except test high temperature shall be + 115° C.

Thermal shock: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202.

No measurements at high humidity. Insulation resistance shall be of at least 200 Megohms within 5 minutes after removal from humidity.

Corona level: Not applicable

RF high potential withstanding voltage: 1500 volts RMS
Frequency - 5-7 MHz.
Leakage current - Not applicable.

Cable retention force: 200 pounds minimum

Torque: 55 inch ounces

Coupling mechanism retention force: 100 pounds min.

RF leakage : -90 dB Tested at a frequency of 2-3 GHz

RF insertion loss: dB max - 0.10 dB max at 4.0 GHz

Preparing activity
NAVY-EC
(Project number 5935-N169-6)

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